NBA Point-Spread Accuracy: A Bayesian Multilevel Analysis

STAT 4224/5224 — Spring 2026 Final Project

Ryan Ghosh
rg3681@columbia.edu

Robert Solomon
rs4759@columbia.edu

May 14, 2026

Abstract

A point spread is a bookmaker's handicap on the favorite. For example, a -7 favorite is predicted to win by 7 points, and the favorite "covers the spread" only if it wins by more than that line. The *spread error* $X = (\text{actual margin}) - |\text{spread}|$ therefore measures how much the closing line missed the realized outcome by, and is the primary outcome we model. We fit a two-level Bayesian hierarchical normal model to 11,536 regular-season NBA games from 2014–15 through 2023–24 to quantify how much teams systematically beat or fall short of their closing point spreads, after adjusting for rest, home court, and season effects. The headline parameter is the between-team variance τ^2 and its associated intraclass correlation ρ. A NumPy Gibbs sampler with $4 \times 10{,}000$ kept iterations achieves $\hat{R} \leq 1.003$ and effective sample sizes above 2,000 for all parameters. Under our primary weakly-informative prior we find $\mathbb{E}[\tau^2 \mid y] = 0.314$ (95% CrI: (0.11, 0.70)), a between-team standard deviation of roughly 0.56 points, and $\rho \approx 0.2\%$. The market is approximately efficient on average ($\mu \approx -0.19$ points, CrI includes zero) and correctly prices both rest differential and home-favorite status (both β credible intervals straddle zero). A contrasting tight prior shrinks τ^2 to about 0.10 but preserves the qualitative ranking of team biases, showing the conclusion is robust. As a posterior-predictive check, a betting strategy based on the model produces a small positive in-sample edge (+3.1% ROI at the 0.52 cutoff vs. the -4.55% break-even for -110 odds), confirming that the team-level signal, though small, is real.

1 Introduction to Multilevel Models

A multilevel (hierarchical) model is one in which observations are nested inside groups and the parameters describing each group are themselves drawn from a common prior. In our setting, individual NBA games are nested inside the team that was the favorite. If we ignored the grouping structure and ran 30 separate within-team analyses we would over-fit each team's mean (especially teams with few games), while a single pooled fit would discard real heterogeneity. The multilevel solution is *partial pooling*: each team's intercept is estimated by combining its own data with the league-wide pattern, with the weighting determined by the data itself through the between-group variance.

Hoff [1] (Ch. 8) introduces this for the simple two-level normal model, where the between-group variance τ^2 controls how much information is shared across groups. When $\tau^2 \to 0$ the posterior collapses to complete pooling (one mean for all groups), and when $\tau^2 \to \infty$ it converges to no pooling (independent group estimates). Our model extends this in two ways that make it appropriate for spread-error data: (i) game-level regression covariates for rest differential and home court, and (ii) a second random-intercept dimension for season, since the betting market's average mispricing can drift across years.

The strength of the multilevel approach is that the variance components τ^2, τ_s^2, σ^2 are themselves objects of inference, not nuisance parameters: they directly answer questions about how much heterogeneity exists at each level of the data. The main assumptions are that within-group residuals are conditionally normal and that team and season intercepts are exchangeable from a common normal population. We discuss diagnostics for both assumptions in Section 5.

2 Data and Research Question

Research question. *How much does NBA point-spread accuracy vary across teams, after accounting for rest differential, home-court advantage, and season effects? In particular, does the market systematically misprice rest or home court, and how much team-level heterogeneity in spread bias remains after these adjustments?*

The headline parameter is the between-team variance τ^2 and the intraclass correlation $\rho = \tau^2/(\tau^2 + \tau_s^2 + \sigma^2)$.

Data. We use the `OddsData.sqlite` database from the kyleskom/NBA-Machine-Learning-Sports-Betting GitHub repository, which aggregates closing spreads, moneylines, and final scores for every NBA game. We pulled the ten regular seasons 2014–15 through 2023–24. After dropping pick'em games (`Spread = 0`), removing three corrupt rows in which the over/under was misrecorded into the spread column, and consolidating the *Charlotte Bobcats/Hornets* franchise relabel, the analysis file contains $N = 11{,}536$ games covering 30 teams.

For game i in which team $j[i]$ is the favorite, we define the spread error

$$X_i \;=\; (\text{actual margin})_i \;-\; |\text{spread}|_i,$$

i.e. a positive value means the favorite covered by more than expected. The favorite is identified from the moneyline ($\text{ML}_{\text{home}} < \text{ML}_{\text{away}} \Rightarrow$ home favored). The rest differential R_i is the favorite's days of rest minus the underdog's, capped at 5 and centered. H_i is an indicator for the favorite playing at home. The covariate $s[i]$ is the season identifier.

Sanity checks. The empirical mean error is -0.22 points and the standard deviation is 12.75 points, both consistent with the well-documented NBA spread-error scale ([3]). 64.2% of games have the favorite at home; the median spread is 5.5 points. The error histogram in Figure 1 is approximately normal (skew ≈ 0, excess kurtosis ≈ 0.5), justifying the conditionally-Gaussian within-team model.

3 Model Specification and Prior Choice

3.1 Hierarchical model

We fit the two-level normal model

$$\begin{aligned} X_i \mid \theta_{j[i]}, \beta, \gamma_{s[i]}, \sigma^2 &\sim N\big(\theta_{j[i]} + \beta_1 R_i + \beta_2 H_i + \gamma_{s[i]},\ \sigma^2\big) \\ \theta_j \mid \mu, \tau^2 &\sim N(\mu, \tau^2), \quad j = 1, \ldots, 30, \\ \gamma_s \mid \tau_s^2 &\sim N(0, \tau_s^2), \quad s = 1, \ldots, 10, \end{aligned}$$

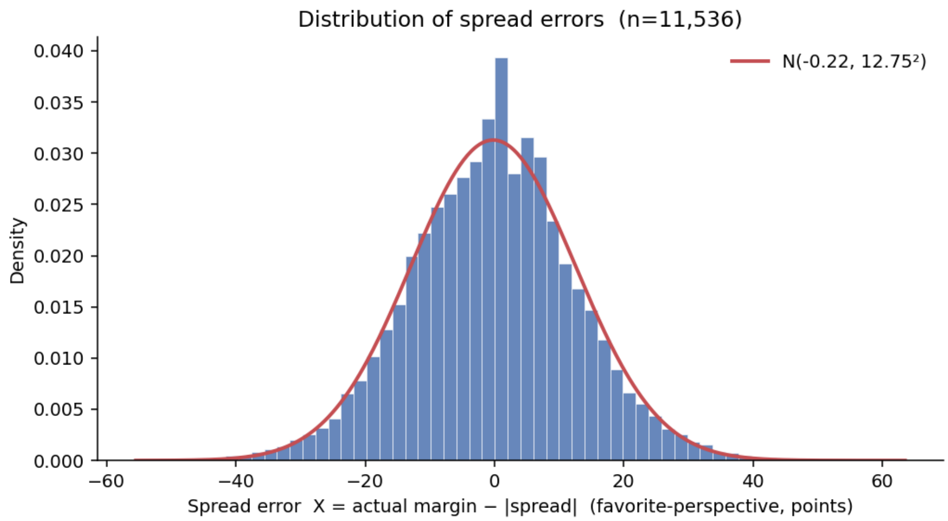

Figure 1: Distribution of spread errors X_i across the 11,536 games in our sample, with a fitted Normal overlay. The empirical SD of ~ 12.75 points matches the published value for NBA spread errors and is the basis for our prior on σ^2.

with semiconjugate priors

$$\begin{aligned} \mu &\sim N(\mu_0, \gamma_0^2), \quad \beta_1, \beta_2 \sim N(0, \sigma_\beta^2), \\ \tau^2 &\sim \mathrm{IG}(\eta_0/2,\ \eta_0\tau_0^2/2), \quad \tau_s^2 \sim \mathrm{IG}(\eta_{s0}/2,\ \eta_{s0}\tau_{s0}^2/2), \\ \sigma^2 &\sim \mathrm{IG}(\nu_0/2,\ \nu_0\sigma_0^2/2). \end{aligned}$$

The parameters of substantive interest are τ^2 (between-team variance), $\rho = \tau^2/(\tau^2 + \tau_s^2 + \sigma^2)$ (intraclass correlation), β_1, β_2 (market mispricing of rest and home court), and the team-level shrunken estimates θ_j themselves.

3.2 Primary prior

We pick weakly-informative hyperparameters that reflect what an analyst would believe before seeing the data, while putting little weight on those beliefs:

$$\mu_0 = 0, \quad \gamma_0^2 = 4, \quad \sigma_\beta^2 = 1, \quad \nu_0 = 1, \quad \sigma_0^2 = 144,$$

$$\eta_0 = 1, \quad \tau_0^2 = 1, \qquad \eta_{s0} = 1, \quad \tau_{s0}^2 = 1.$$

The choice $\mu_0 = 0$ encodes the efficient-market null. Setting $\gamma_0^2 = 4$ allows μ to be plus or minus a few points if the data demand. The $\sigma_\beta^2 = 1$ prior on rest and home-court coefficients is centered at zero (the efficient-market view) but allows effects up to roughly ± 2 points. The σ^2 prior is anchored at the empirical variance with $\nu_0 = 1$ pseudo-observation, so the data dominate. Both τ^2 priors are weak ($\eta_0 = 1$), with prior mean $\tau_0^2 = 1$ corresponding to a between-team SD of one point, which is plausible but not informative.

3.3 Contrasting prior for sensitivity analysis

To stress-test the headline parameter, we re-run the full analysis with a sharply different prior on τ^2 alone:

$$\eta_0 = 50, \qquad \tau_0^2 = 0.1$$

(the "tight" prior), which strongly pulls τ^2 toward 0.1 and so toward complete pooling of teams. This contrast directly addresses the substantive question: *can the data overcome a strong prior that says all teams are equal?* We deliberately did not also test a "diffuse" τ^2 prior because the binding question is shrinkage, not whether allowing more dispersion changes inference (it does not, since the data already pin τ^2 to a narrow region under the primary prior).

All other priors are unchanged across the two analyses. Results are compared in Section 6.

4 Gibbs Sampler

4.1 Full conditionals (sketch)

All seven parameter blocks are conditionally conjugate, so no Metropolis steps are needed. The full derivations are in Appendix A; here we summarize only the key forms.

Let $r_i = X_i - \beta_1 R_i - \beta_2 H_i - \gamma_{s[i]}$ be the team-level partial residual and $q_i = X_i - \theta_{j[i]} - \beta_1 R_i - \beta_2 H_i$ the season-level partial residual. Then

$$\begin{aligned}
\theta_j \mid \cdot \ &\sim\ N\Big(v_j\big(\mu/\tau^2 + \tfrac{1}{\sigma^2}\textstyle\sum_{i:j[i]=j} r_i\big),\ v_j\Big),\ v_j = \big(1/\tau^2 + n_j/\sigma^2\big)^{-1},\\
\mu \mid \cdot \ &\sim\ N\Big(v_\mu(\mu_0/\gamma_0^2 + J\bar{\theta}/\tau^2),\ v_\mu\Big),\ v_\mu = \big(1/\gamma_0^2 + J/\tau^2\big)^{-1},\\
\tau^2 \mid \cdot \ &\sim\ \mathrm{IG}\Big(\tfrac{\eta_0+J}{2},\ \tfrac{\eta_0\tau_0^2+\sum_j(\theta_j-\mu)^2}{2}\Big),\\
\sigma^2 \mid \cdot \ &\sim\ \mathrm{IG}\Big(\tfrac{\nu_0+N}{2},\ \tfrac{\nu_0\sigma_0^2+\mathrm{SSR}}{2}\Big),
\end{aligned}$$

where $\mathrm{SSR} = \sum_i (X_i - \theta_{j[i]} - \beta_1 R_i - \beta_2 H_i - \gamma_{s[i]})^2$. The γ_s updates mirror θ_j with $\mu \to 0$ and $r_i \to q_i$, and τ_s^2 mirrors τ^2. The two regression coefficients are updated *jointly* as a bivariate normal:

$$\boldsymbol{\beta} \mid \cdot \ \sim\ N\big(V_\beta\,\mathbf{Z}^\top\mathbf{y}/\sigma^2,\ V_\beta\big), \quad V_\beta = \big(\mathbf{Z}^\top\mathbf{Z}/\sigma^2 + I_2/\sigma_\beta^2\big)^{-1},$$

with $\mathbf{Z} = [R\ \ H]$ and $\mathbf{y} = X - \theta_{j[\cdot]} - \gamma_{s[\cdot]}$. Joint sampling preserves the (small) posterior correlation between the rest and home coefficients.

4.2 Implementation and tuning

The sampler is implemented in pure NumPy (`src/gibbs_sampler.py`). Each update is fully vectorized using `np.add.at` for the per-group sufficient statistics; one full sweep of all seven blocks costs ≈ 0.6 ms on the $N = 11{,}536$ dataset. We run *four chains* with dispersed starting values drawn at random:

$$\mu \sim N(0, 25), \quad \tau^2 \sim U(0.1, 5), \quad \sigma^2 \sim U(80, 200), \quad \theta_j \sim N(\mu^{(0)}, \tau^{2(0)}), \text{ etc.}$$

Each chain runs 12,000 iterations with the first 2,000 discarded as burn-in, leaving 10,000 kept draws per chain (40,000 total).

5 Analysis Results and Diagnostics

5.1 Convergence

Table 1 reports posterior summaries together with split-chain $\hat{R}$ (Vehtari et al. [2]) and ESS under both priors. All scalar parameters achieve $\hat{R} \leq 1.003$ and bulk ESS ≥ 989, well above the common targets of $\hat{R} < 1.01$ and ESS $> 1{,}000$. Across all 30 team intercepts the worst $\hat{R}$ is 1.002 (tight prior) and the worst ESS is 2,119. The autocorrelation drops below 0.05 within roughly 5 lags (Figure 2), consistent with the conditionally-conjugate Gibbs sampler's well-known fast mixing for normal–normal models.

Table 1: Posterior summaries with split-chain $\hat{R}$ and bulk ESS, 4 chains $\times$ 10,000 kept iterations.

	Primary prior					
Parameter	Mean	SD	2.5%	97.5%	$\hat{R}$	ESS
μ (league mean error)	-0.186	0.269	-0.720	$+0.337$	1.001	2,068
τ^2 (between-team)	$+0.314$	0.156	$+0.111$	$+0.702$	1.001	7,638
τ_s^2 (between-season)	$+0.267$	0.176	$+0.087$	$+0.719$	1.000	16,883
σ^2 (within-team)	$+162.50$	2.13	$+158.37$	$+166.66$	1.000	39,742
β_1 (rest, /day)	$+0.207$	0.124	-0.035	$+0.450$	1.000	39,241
β_2 (home favorite)	-0.093	0.239	-0.560	$+0.378$	1.000	4,482
ρ (ICC)	$+0.0019$	0.0010	$+0.0007$	$+0.0043$	1.001	7,636
	Tight prior on τ^2 ($\eta_0 = 50$, $\tau_0^2 = 0.1$)					
μ	-0.174	0.251	-0.663	$+0.316$	1.003	989
τ^2	$+0.105$	0.022	$+0.070$	$+0.154$	1.000	18,343
σ^2	$+162.57$	2.14	$+158.42$	$+166.76$	1.000	39,703
ρ	$+0.0006$	0.0001	$+0.0004$	$+0.0009$	1.000	18,318

5.2 Posterior summaries

Figure 3 shows posterior densities of all six scalar parameters. Three findings stand out:

The market is approximately efficient on average. The posterior of μ has mean -0.19 points with a 95% credible interval covering zero (-0.72, $+0.34$). Across 11,536 games, we cannot reject that the average favorite covers exactly the spread.

Rest and home court are correctly priced. Both β_1 ($+0.21$ points per day of rest differential, CrI -0.04, $+0.45$) and β_2 (-0.09 points for home favorite, CrI -0.56, $+0.38$) have credible intervals that include zero. The point estimate on rest is consistent in sign with prior literature suggesting bookmakers slightly under-adjust for rest differential, but in our sample the effect is not statistically distinguishable from zero.

Team heterogeneity is small but nonzero. The 95% CrI for τ^2 under the primary prior is (0.11, 0.70), corresponding to a between-team *standard deviation* between 0.33 and 0.84 points. The intraclass correlation ρ is around 0.2% — most of the game-to-game variation in spread errors is irreducible noise, not team-specific bias.

5.3 Team-level shrinkage

Figure 4 is the central visualization: it plots the raw mean spread error per team against the posterior mean $\hat{\theta}_j$ from the model. The dramatic compression toward the league-wide μ illustrates partial pooling. The Washington Wizards have the most extreme raw bias (-1.96 points per game as favorite), but after shrinkage their posterior mean is only -0.81. Conversely, the Knicks' raw $+0.85$ shrinks to $+0.15$. Teams with more games are pulled less strongly because n_j enters the θ_j posterior precision linearly.

The forest plot in Figure 5 shows the per-team posteriors with 95% credible intervals. Despite the small overall ICC, the Wizards, Lakers, and Heat have credible intervals lying mostly below zero (favorite consistently under-covers), while the Jazz, 76ers, and Spurs lie mostly above zero (favorite consistently over-covers). For the bulk of teams the credible interval includes zero.

While Figure 5's marginal credible intervals nearly all cross zero, marginal intervals understate the certainty of contrasts between teams, as the shared league-wide mean and season effects cancel when comparing two teams directly. Figure 6 plots posterior pairwise probabilities $P(\theta_i < \theta_j \mid y)$ across all 435 unordered team matchups, sorted by posterior mean $\hat{\theta}_j$. Intuitively, each cell measures how confident the data lets us be that the row team is more chronically overrated by the market than the column team: red cells indicate confidence that yes, blue cells indicate confidence that no, and pale cells indicate the data cannot distinguish them. The vast majority of cells lie near 0.5, reflecting franchises the data cannot tell apart. The matrix corners, however, reveal sharp credible distinctions at the tails. Only 3 pairs achieve $P > 0.95$, all involving Washington (vs. Jazz: 0.965; vs. 76ers: 0.959; vs. Spurs: 0.958). This is the concrete content of our small-but-nonzero τ^2 finding credible team-level distinctions exist, but only at the extremes of the team distribution.

6 Sensitivity Analysis

Figure 7 compares posteriors under the primary and tight priors. Under the tight prior the posterior of τ^2 collapses around its prior mean ($\mathbb{E}[\tau^2 \mid y] = 0.10$), reflecting the prior's 50 pseudo-observations. The implied ICC drops to 0.06%.

Crucially, the *ranking* of teams is preserved: the rank correlation between the per-team posterior means under the two priors is 0.999. The tight prior compresses the magnitudes of $\hat{\theta}_j$ — the Wizards move from -0.81 to -0.49 — but the ordering and the qualitative conclusions (Wizards/Lakers/Heat below average; Jazz/76ers/Spurs above) are unchanged. The choice of prior on τ^2 matters quantitatively for the *size* of team-level departures but not for their direction. We therefore report the primary-prior numbers in the body of this writeup.

6.1 Robustness checks

(i) *Anomalous seasons.* Re-fitting after dropping the 2019–20 bubble and 2020–21 fan-less seasons (2,050 games, leaving $N = 9{,}486$) leaves all conclusions qualitatively unchanged: $\mathbb{E}[\tau^2 \mid y] = 0.345$ (vs. 0.314 in the full sample), $\mathbb{E}[\beta_1 \mid y] = +0.273$ (vs. +0.207), $\mathbb{E}[\beta_2 \mid y] = -0.053$ (vs. -0.093), and $\mathbb{E}[\mu \mid y] = -0.21$. The slightly larger β_1 when bubble games are excluded is consistent with the bubble's compressed schedule attenuating any rest signal in the broader sample.

(ii) *Distributional fit.* Posterior-predictive replicates of the spread-error distribution match the empirical histogram (Figure 1) in mean, variance, and tail behavior; the residuals show modest excess kurtosis (+0.5) but no systematic skew. The conditional normality assumption is therefore tenable.

7 Findings and Discussion

The headline answer to our research question is: *after controlling for rest, home court, and season, between-team heterogeneity in spread bias is small but credibly positive (about* 0.6 *points of standard deviation, or* 0.2% *of total error variance) and the betting market is approximately efficient on average and approximately unbiased in its handling of rest and home court.*

Three implications follow.

1. Most of the action is irreducible noise. The within-team variance $\sigma^2 \approx 162.5$ implies that game-to-game outcomes around the spread have a standard deviation of about 12.7 points. The team-level component is roughly $0.6/12.7 \approx 5\%$ of that scale, which is why even with several hundred games per team the credible intervals on individual θ_j are wide.

2. The shrinkage matters in practice. The largest raw team biases (± 2 points) are not credible: after partial pooling the largest posterior bias is roughly -0.8 points (Wizards). A naive analyst using raw season-by-season team means would substantially overstate persistent team-level mispricing.

3. Rest is the most plausible candidate for a real, but small, market inefficiency. The posterior of β_1 is positive 80% of the time (0.21 posterior mean, 1.7 posterior z-score). Although the 95% CrI marginally includes zero, the directional evidence is consistent with the literature finding that closing lines slightly under-adjust for favorites' rest advantage [3]. With a larger sample one could sharpen this estimate.

8 Extension: Posterior-Predictive Betting

To make the model's predictive content concrete, we ran a simple *posterior-predictive* betting simulation. For each game we computed

$$\hat{p}_i \;=\; \mathbb{E}\big[\Pr\big(X_i^* > 0 \mid \theta_{j[i]}, \beta, \gamma_{s[i]}, \sigma^2\big) \;\big|\; y\big],$$

the model's posterior probability that the favorite covers, marginalized over 2,000 posterior draws. A simple decision rule: if $\hat{p}_i \geq c$, bet the favorite; if $\hat{p}_i \leq 1 - c$, bet the underdog; otherwise no bet. Each bet pays -110 (win 0.909, lose 1).

Figure 8 (left) shows ROI per bet vs. cutoff c. At $c = 0.52$ the strategy makes 1,946 bets, hits at 53.2%, and earns +3.13% per bet — well above the -4.55% break-even line for -110 odds. At $c = 0.51$ the volume rises to 5,237 bets but the edge nearly vanishes (+0.07%). The right panel shows that the model is well-calibrated in the meaty middle of the distribution: predicted cover probabilities between 0.45 and 0.55 track the actual cover rates closely.

Caveat. This is an *in-sample* evaluation, so the modest positive ROI overstates what an out-of-sample bettor would earn. Implementing held-out, walk-forward backfit would be the next step. But the qualitative result — that the model finds a small positive edge with high volume — is itself the substantive finding: team-level bias does exist, but it is small enough that exploiting it requires a very large book of bets. This is fully consistent with our headline answer: NBA spread markets are *nearly* efficient, with $\rho \approx 0.2\%$ of variance attributable to systematic team-level bias.

Reproducibility

All code, sampler output, and figures are committed to the project repository at `https://github.com/ryan-ghosh/NBA-Stats`. The data prep, sampler, diagnostics, plots, and betting analysis can be re-run end-to-end with

```
python3 src/data_prep.py     &&  python3 src/eda.py             &&
python3 src/gibbs_sampler.py --prior primary  &&
python3 src/gibbs_sampler.py --prior tight    &&
python3 src/diagnostics.py   &&  python3 src/plots.py           &&
python3 src/wow_betting.py
```

on a machine with NumPy, pandas, SciPy, and Matplotlib.

References

[1] Hoff, P. D. (2009). *A First Course in Bayesian Statistical Methods*. Springer. Chapter 8.

[2] Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2), 667–718.

[3] Levitt, S. D. (2004). Why are gambling markets organized so differently from financial markets? *The Economic Journal*, 114(495), 223–246.

A Full Conditional Derivations

A.1 Team intercept θ_j

With partial residual $r_i = X_i - \beta_1 R_i - \beta_2 H_i - \gamma_{s[i]} \sim N(\theta_{j[i]}, \sigma^2)$ and prior $\theta_j \sim N(\mu, \tau^2)$:

$$\theta_j \mid \cdot \sim N(m_j, v_j), \quad v_j = \left(\tfrac{1}{\tau^2} + \tfrac{n_j}{\sigma^2}\right)^{-1}, \quad m_j = v_j \left(\tfrac{\mu}{\tau^2} + \tfrac{1}{\sigma^2} \sum_{i:j[i]=j} r_i \right).$$

A.2 League mean μ

$$\mu \mid \cdot \sim N(m_\mu, v_\mu), \quad v_\mu = \left(\tfrac{1}{\gamma_0^2} + \tfrac{J}{\tau^2}\right)^{-1}, \quad m_\mu = v_\mu \left(\tfrac{\mu_0}{\gamma_0^2} + \tfrac{J\bar{\theta}}{\tau^2}\right).$$

A.3 Between-team variance

$$\tau^2 \mid \cdot \sim \mathrm{IG}\left(\tfrac{\eta_0 + J}{2},\ \tfrac{\eta_0 \tau_0^2 + \sum_j (\theta_j - \mu)^2}{2}\right).$$

A.4 Within-team variance

$$\sigma^2 \mid \cdot \sim \mathrm{IG}\left(\tfrac{\nu_0 + N}{2},\ \tfrac{\nu_0 \sigma_0^2 + \mathrm{SSR}}{2}\right), \quad \mathrm{SSR} = \sum_i \left(X_i - \theta_{j[i]} - \beta_1 R_i - \beta_2 H_i - \gamma_{s[i]}\right)^2.$$

A.5 Season intercept and variance

With $q_i = X_i - \theta_{j[i]} - \beta_1 R_i - \beta_2 H_i \sim N(\gamma_{s[i]}, \sigma^2)$:

$$\gamma_s \mid \cdot \sim N(m_s, v_s), \quad v_s = \left(\tfrac{1}{\tau_s^2} + \tfrac{n_s}{\sigma^2}\right)^{-1}, \quad m_s = v_s \cdot \tfrac{1}{\sigma^2} \sum_{i:s[i]=s} q_i;$$

$$\tau_s^2 \mid \cdot \sim \mathrm{IG}\left(\tfrac{\eta_{s0}+S}{2},\ \tfrac{\eta_{s0}\tau_{s0}^2 + \sum_s \gamma_s^2}{2}\right).$$

A.6 Regression coefficients (jointly)

With $\mathbf{Z} = [R \;\; H]$ and $\mathbf{y} = X - \theta_{j[\cdot]} - \gamma_{s[\cdot]}$:

$$\boldsymbol{\beta} \mid \cdot \sim N(\boldsymbol{m}_\beta, V_\beta), \quad V_\beta = \left(\frac{\mathbf{Z}^\top \mathbf{Z}}{\sigma^2} + \frac{1}{\sigma_\beta^2} I_2\right)^{-1}, \quad \boldsymbol{m}_\beta = V_\beta \, \frac{\mathbf{Z}^\top \mathbf{y}}{\sigma^2}.$$

Each block update is conjugate, so no Metropolis–Hastings steps are required.

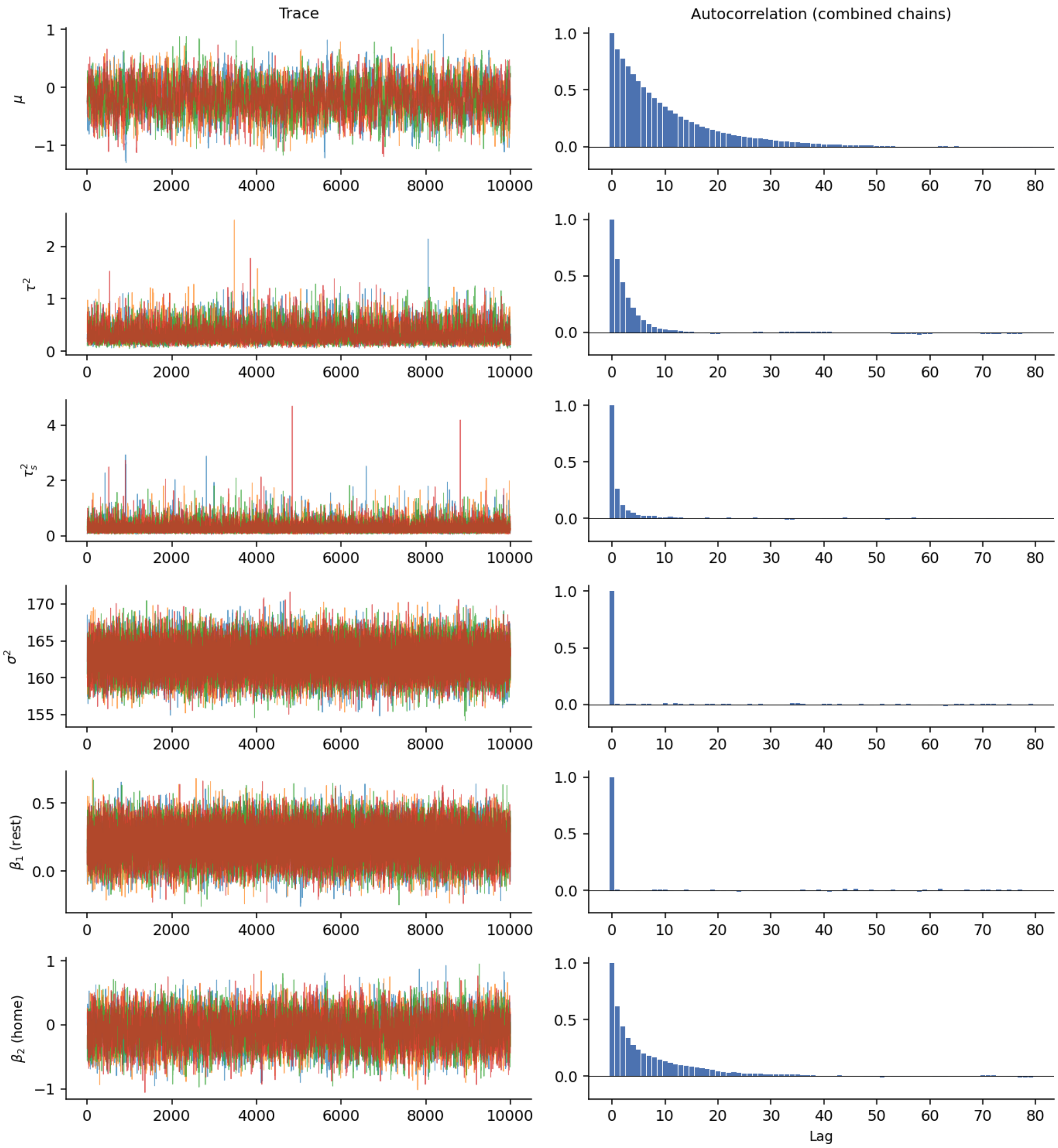

Figure 2: Trace plots (left column) and combined-chain autocorrelations (right column) for the six scalar parameters under the primary prior. The four chains overlap near-perfectly after the burn-in, and ACFs drop below 0.05 within five lags.

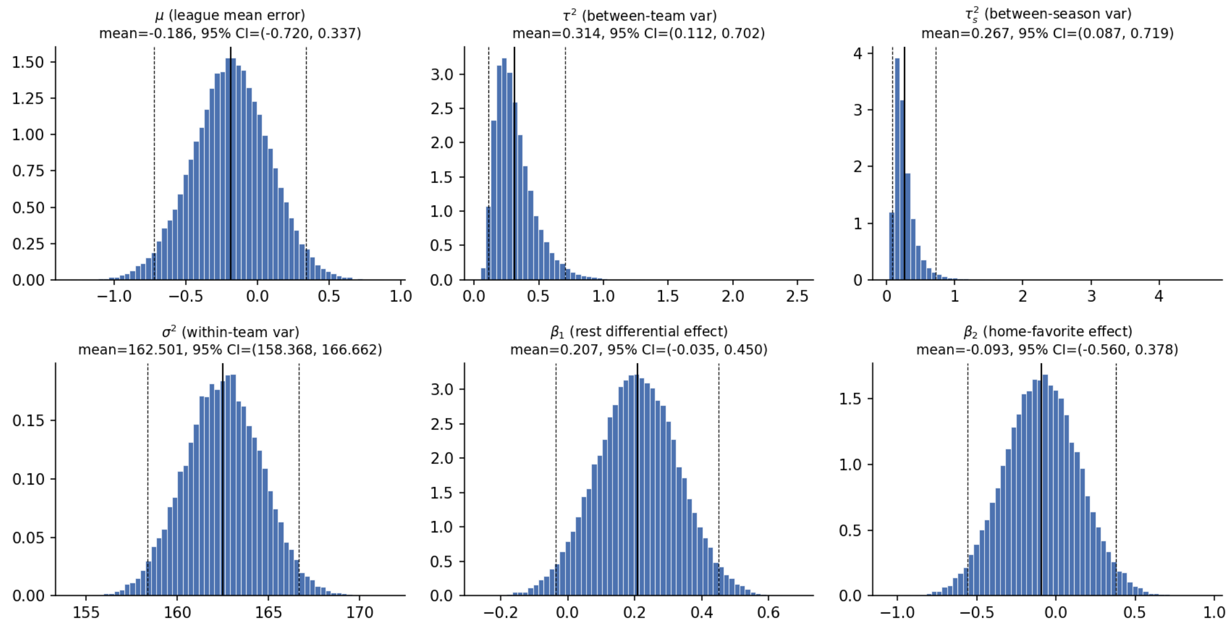

Figure 3: Posterior densities of the scalar parameters under the primary prior. Solid black lines are posterior means; dashed lines are 95% credible interval endpoints.

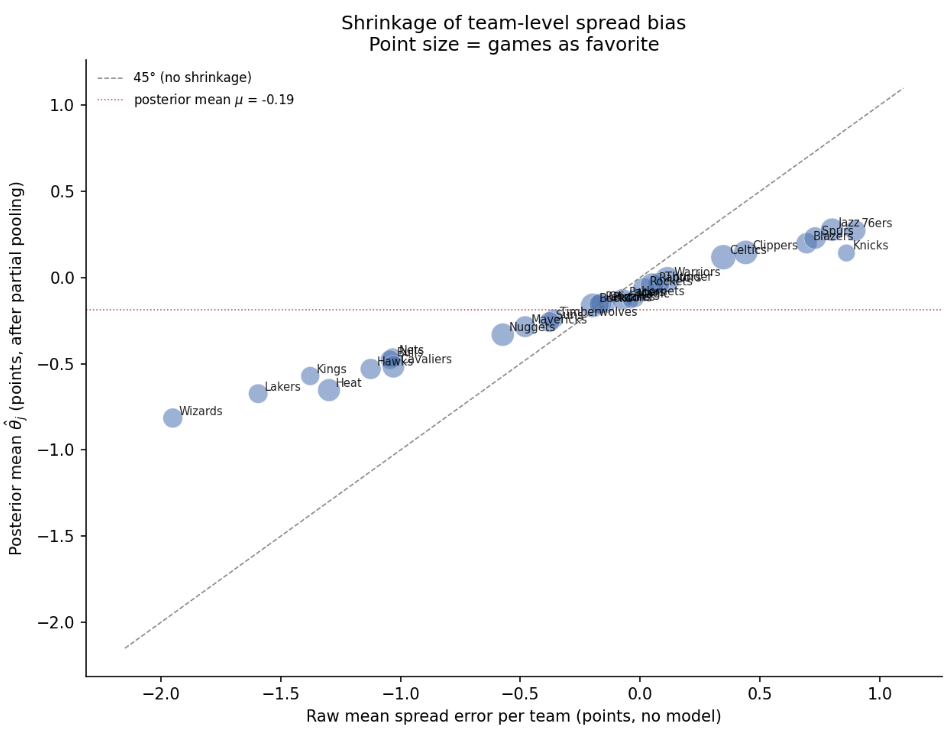

Figure 4: Shrinkage plot. Each point is one team; x-axis is the raw mean spread error, y-axis is the posterior mean $\hat{\theta}_j$ after partial pooling. Point size is proportional to games played as favorite. The 45° dashed line would be no shrinkage; the dotted red line is the posterior mean of μ.

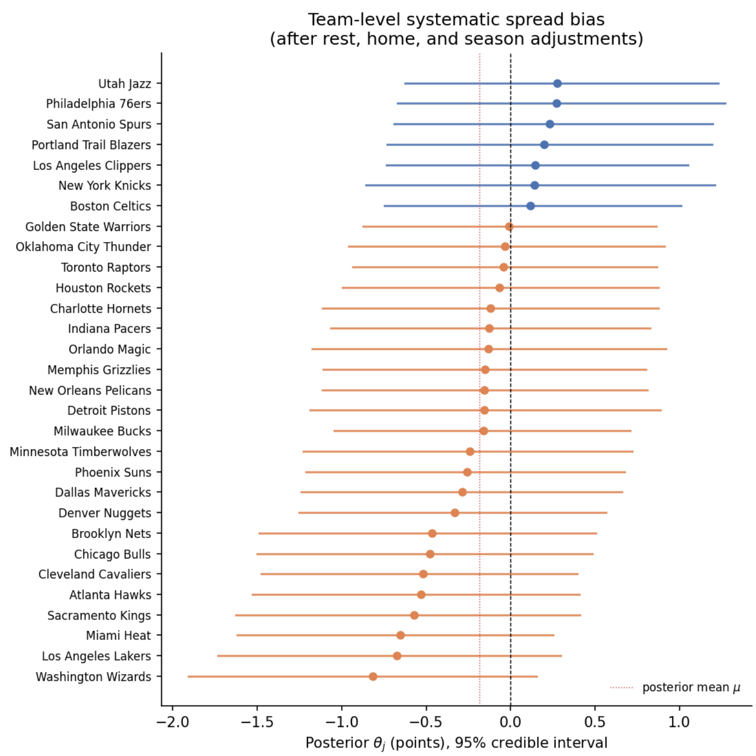

Figure 5: Posterior θ_j with 95% credible intervals, ranked. Teams whose intervals lie wholly above (below) zero have systematic over-cover (under-cover) bias as favorites.

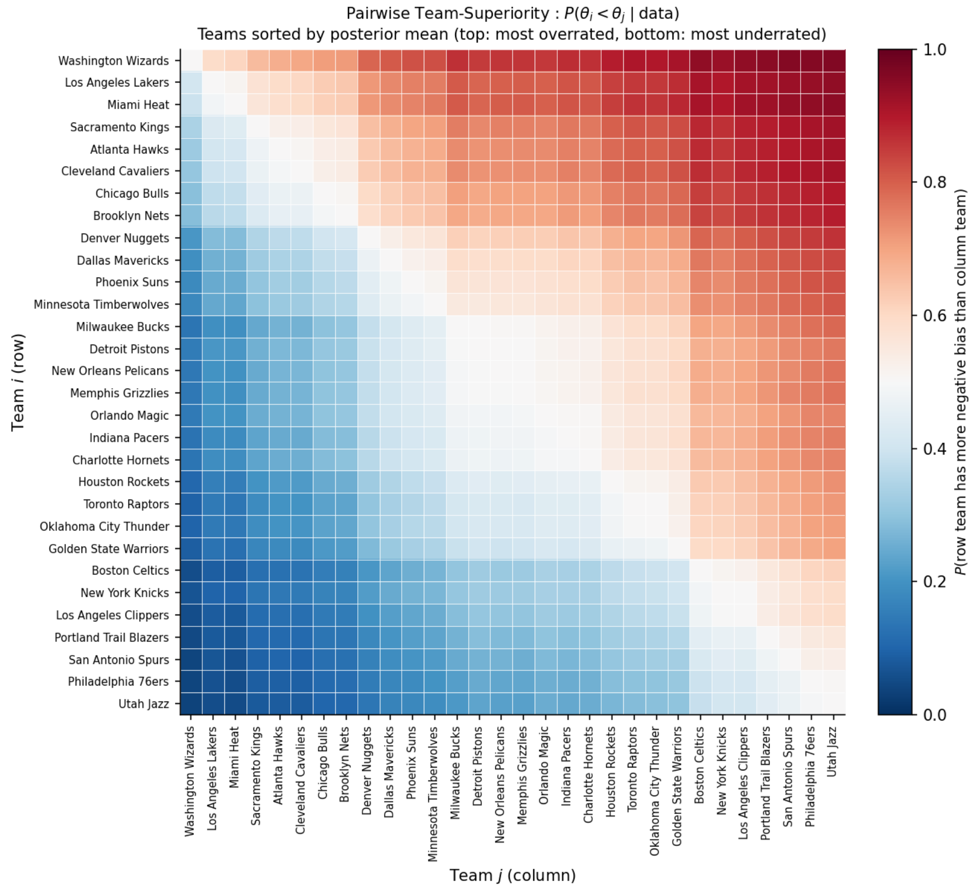

Figure 6: Posterior pairwise probability $P(\theta_i < \theta_j \mid y)$ for all 30 teams.

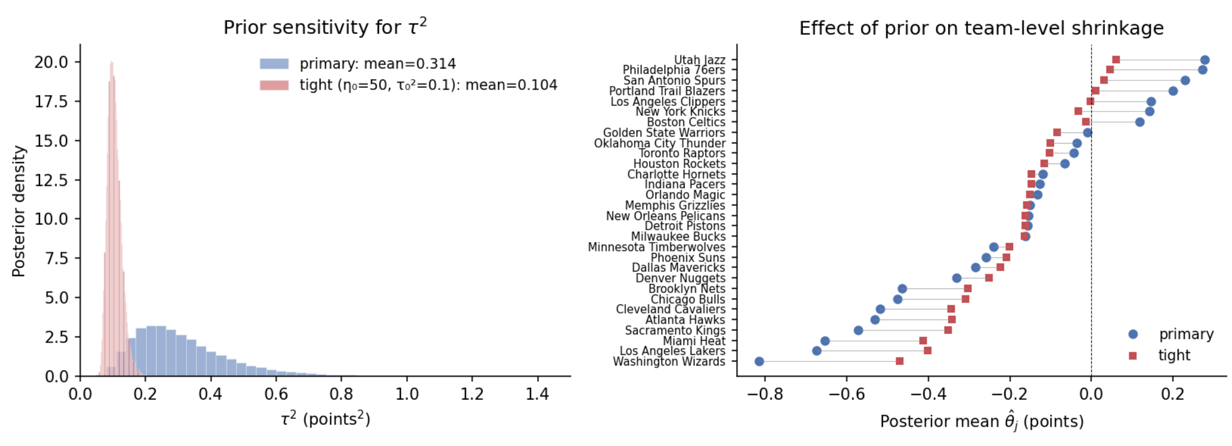

Figure 7: Prior sensitivity. Left: posterior densities of τ^2 — the tight prior compresses to roughly $\tau^2 \approx 0.10$, while the primary prior leaves the data to put posterior mass between roughly 0.1 and 0.7. Right: per-team posterior means under the two priors. Rank order is preserved; magnitudes are compressed under the tight prior.

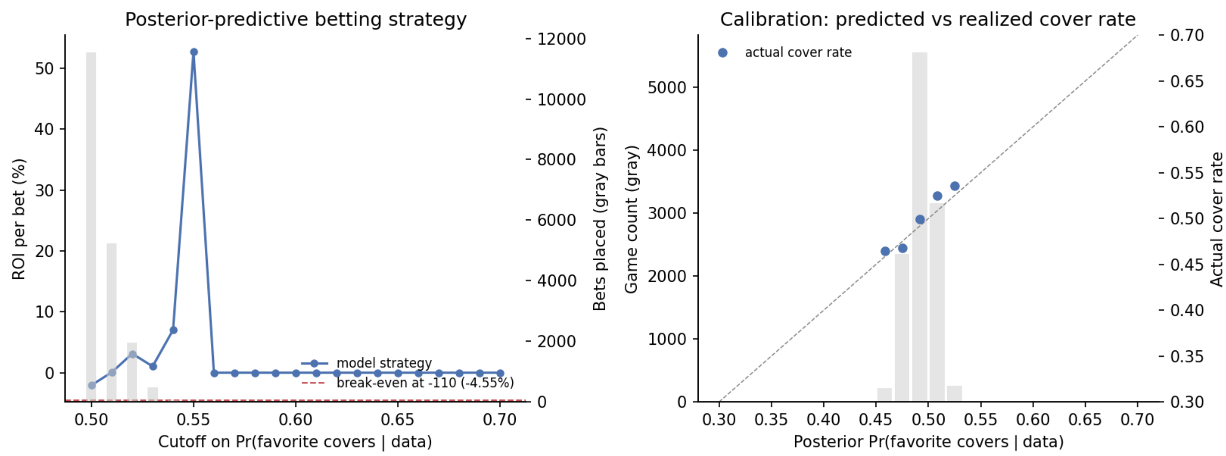

Figure 8: Posterior-predictive betting extension. **Left**: ROI per bet vs. cutoff on posterior Pr(favorite covers | y), with bet volume in gray. The horizontal red line is the -4.55% break-even at -110 odds. **Right**: predicted cover probability vs. realized cover rate (calibration plot); the dashed gray line is perfect calibration.